STATISTICAL FIELD THEORY

The Ising Model

Consider a d-dimensional lattice with N sites $i := \{1, \dots, N\}$, each of which in one of two states $s_i = \pm 1$ (spin up/spin down). This collection has energy

$$E = -\underbrace{B}_{\text{external magnetic field}} \sum_i s_i - J \underbrace{\sum_{i \sim j}}_{\text{neighbouring } i,j} s_i s_j .$$

if $J > 0$, neighbouring states prefer to be aligned $\to$ ferromagnet

Recall that in the canonical ensemble,

probability that i is in state s_i $\forall i$

$$p[s_i] = \frac{e^{-\beta E[s_i]}}{Z},$$

where $k_B = 1$, $\beta = \frac{1}{T}$, and the partition function

$$Z(T, J, B) = \sum_{\{s_i\}} e^{-\beta E[s_i]} .$$

Recall the free energy

$$F_{\text{thermo}}(T, B) = \langle E \rangle - TS = -T \log Z .$$

We call the average spin of the configuration

$$m = \frac{1}{N} \left\langle \sum_i s_i \right\rangle \in [-1, 1]$$

the magnetisation. Note that

$$m = \frac{1}{N} \sum_{\{s_i\}} \frac{e^{-\beta E[s_i]}}{Z} \sum_i s_i = \frac{1}{N\beta} \frac{\partial \log Z}{\partial B} .$$

Let's write

$$Z = \sum_m \underbrace{\sum_{\{s_i\}|m}}_{\text{sum over } \{s_i\} \text{ s.t. } \frac{1}{N}\sum s_i = m} e^{-\beta E[s_i]} =: \sum_m e^{-\beta F(m)}$$

can depend on T, and hence on β. Hard to calculate

$$\overset{\text{in the limit as } N \to \infty}{=} \frac{N}{2} \int_{-1}^{+1} dm \, e^{-\beta F(m)} .$$

We call $F(m)$ the effective free energy. Let

$$f(m) = \frac{F(m)}{N},$$

so

$$Z = \frac{N}{2} \int_{-1}^{1} dm \, e^{-\beta N f(m)} .$$

We assume that $N \sim 10^{23} \gg 0$ while $\beta N \sim 1$, so this integral is well approximated by the value of the integrand at $m = m_{min}$, minimising f, i.e.

$$\left.\frac{\partial f}{\partial m}\right|_{m = m_0} = 0 .$$

Hence,

$$Z \approx e^{-\beta N f(m_{min})},$$

and $F_{\text{thermo}} \approx F(m_{min})$.

We will try to find $F(m) = N f(m)$ using mean field theory. Some of this is not going to work! To do this, we want to sum over $\{s_i\}$ with $\sum s_i = Nm$. We first just replace each s_i with its mean value m

$$E = -B \sum_i m - J \sum_{i \sim j} m^2 \implies \frac{E}{N} = -Bm - \frac{1}{2} J \underbrace{q}_{\#\text{neighbours}} m^2 .$$

We have

$$m = \frac{N_\uparrow - N_\downarrow}{N} = \frac{2N_\uparrow - N}{N},$$

with the number of such configurations

$$\Omega = \frac{N!}{N_\uparrow! \, (N - N_\uparrow)!} .$$

By Stirling,

$$\frac{\log \Omega}{N} \approx \log 2 - \frac{1}{2}(1+m)\log(1+m) - \frac{1}{2}(1-m)\log(1-m) .$$

With the mean field approximation,

$$e^{-\beta N f(m)} \approx \Omega(m) \, e^{-\beta E(m)},$$

and hence

$$f(m) \approx -Bm - \frac{1}{2} J q m^2 - T\left(\log 2 - \frac{1}{2}(1+m)\log(1+m) - \frac{1}{2}(1-m)\log(1-m)\right).$$

Calculating the minimum of $f(m)$, we get

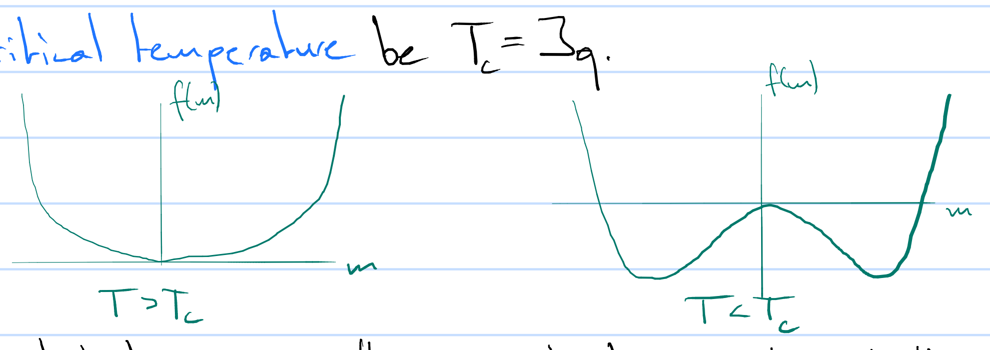
$$m = \tanh(\beta B + \beta J q m),$$

the self-consistency equation. This can also be derived by assuming that each spin experiences an effective magnetic field $B_{\text{eff}} = B + Jqm$, the mean field, thus replacing the tricky interactions.

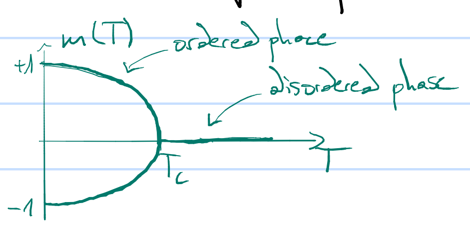
Landau Approach to Phase Transitions

Recall that
$$f(m) \approx -Bm - \frac{1}{2}Jqm^2 - T\left(\log 2 - \frac{1}{2}(1+m)\log(1+m) - \frac{1}{2}(1-m)\log(1-m)\right).$$
For small m,
$$f(m) \approx -T\log 2 - Bm + \frac{1}{2}(T-Jq)m^2 + \frac{1}{12}Tm^4 + \dots$$

Let now $B=0$. We are interested in the locations of the minima of f, so we can drop the constants. Then
$$f(m) \approx \frac{1}{2}(T-Jq)m^2 + \frac{1}{12}Tm^4 + \dots$$
Let the critical temperature be $T_c = Jq$.



We see that at high energies, the magnetisation vanishes at the minimum $m=0$. At energies below T_c, the free energy is minimised at $m = \pm m_0$, where
$$m_0 \approx \sqrt{\frac{3(T_c - T)}{T}}$$
when $T_c \approx T$.



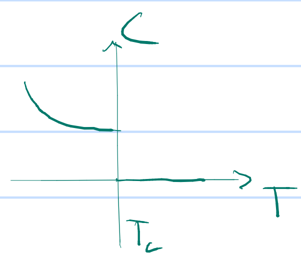
This kind of phase transition is called continuous, or second order phase transition.

Recall the heat capacity $C = \frac{\partial \langle E \rangle}{\partial T}$. In the canonical ensemble, $\langle E \rangle = -\frac{\partial \log Z}{\partial \beta}$, so
$$C = \beta^2 \frac{\partial^2}{\partial \beta^2}\log Z.$$
We still need to compute $\log Z$, i.e. compute $f(m_{min})$. For $T > T_c$, $m_{min} = 0 = f(m_{min})$.

still neglecting the constant term

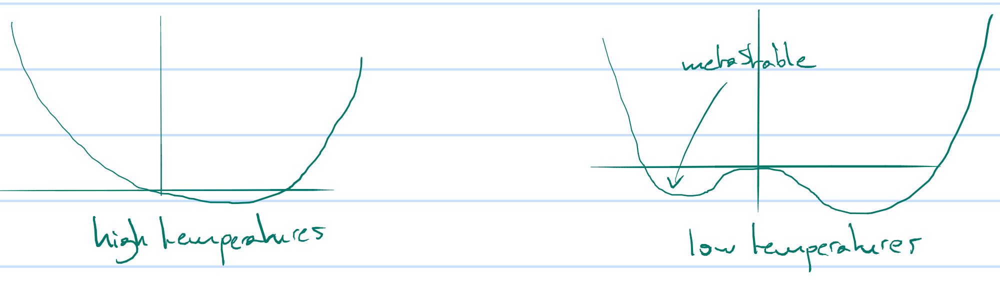
For $T < T_c$, $f(m_{min}) = -\frac{3}{4}(T_c - T)^2/T$.

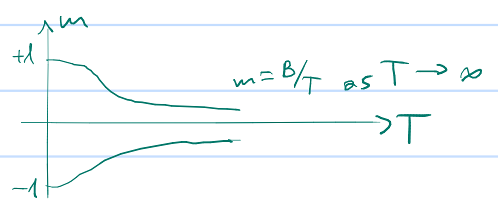
We get
$$c = \frac{C}{N} = \begin{cases} 0 & \text{as } T \to T_c^+ \\ \frac{3}{2} & \text{as } T \to T_c^-, \end{cases}$$



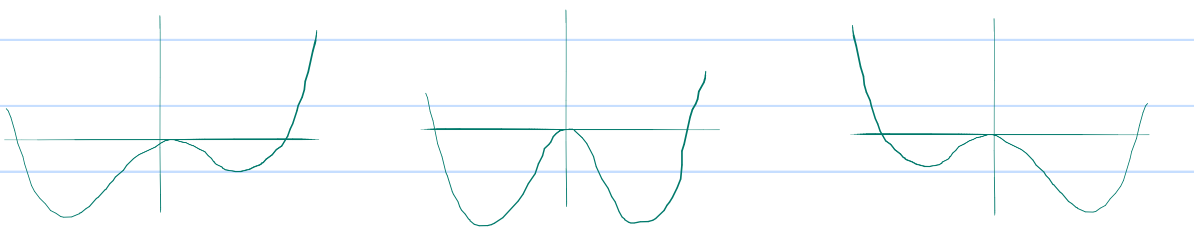
a discontinuous jump.

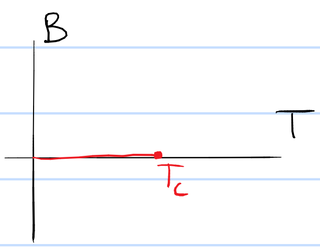
Let now $B \neq 0$. Then, ignoring constants,
$$f(m) \approx -Bm + \frac{1}{2}(T-Jq)m^2 + \frac{1}{12}Tm^4 + \dots$$





Here, there is no phase transition. Even the branch of m is determined by the sign of B. If we fix $T < T_c$ and vary B from negative to positive, we do get a first order phase transition.





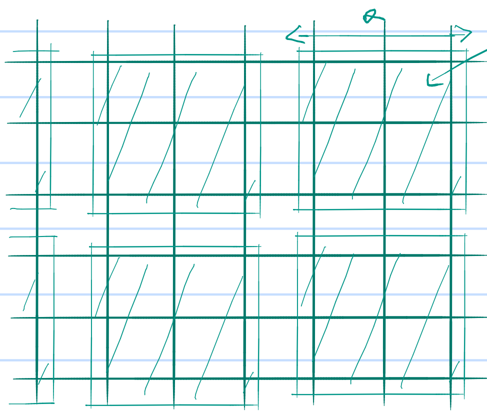
Phase diagram:
Moving through the red line yields a 1st order phase transition, at T_c a 2nd order PT, after none
$\rightsquigarrow (B,T) = (0, T_c)$ is a critical point

Note: It turns out that Mean Field Theory
- fails for 1 dimension (no phase transitions exist),
- predicts the basic structure in 2 and 3 dimensions correctly, but not the details at critical temperature,
- is correct in ≥ 4 dimensions

for the Ising model. Intuitively, this is because of the larger number of neighbours. Generally, mean field theory always works above an upper critical dimension d_c, and fails below a lower critical dimension d_l.

Landau-Ginzburg Theory

The Landau approach misses all spatial variations in the model by reducing the configuration of the Ising model to a single number m. Here, we promote it to a local order parameter $m(x)$. This is done by coarse graining:



N' nodes $\gg 0$

$m(x) = \frac{1}{N'}\sum s_i$

box center

Assumptions:

$\frac{N}{N'} \gg 0 \rightsquigarrow x$ is continuous

$N' \gg 0 \rightsquigarrow m(x)$ is continuous in $[-1,1]$

Landau-Ginzburg free energy functional

We have

$$Z = \sum_{m(x)} \sum_{\{s_i\}|m(x)} e^{-\beta E[s_i]} =: \sum_{m(x)} e^{-\beta F[m(x)]}$$

$$=: \int Dm\, e^{-\beta F[m]}.$$

Note that the Landau-Ginzburg free energy is

- Local: $F[m(x)] = \int d^d x\, f[x]$,

  where f is local, depending on $m(x)$ and its derivatives only.
- Translation and rotation invariant: This follows from the lattice symmetries
- $\mathbb{Z}_2$-symmetric: The Ising model is invariant under $s_i \to -s_i$ with $B \to -B$, i.e. $m(x) \to -m(x)$. Hence, so is F.

We also assume that F is analytic in $m(x)$ and its derivatives. We will also consider lower order derivatives of $m(x)$ more important.

Hence, for $B=0$,

flips sign at 2nd order phase transition $\quad > 0$

$$F[m(x)] = \int d^d x \left[\frac{1}{2}\alpha_2(T) m^2 + \frac{1}{4}\alpha_4(T) m^4 + \frac{1}{2}\gamma(T)(\nabla m)^2 + \dots\right].$$

We will look for minimisers again:

$$\delta F = \int d^d x \left[\alpha_2 m\,\delta m + \alpha_4 m^3 \delta m + \gamma \nabla m \cdot \nabla \delta m\right]$$

$$= \int d^d x \left[\alpha_2 m + \alpha_4 m^3 - \gamma \nabla^2 m\right]\delta m,$$

giving

$$\frac{\delta F}{\delta m}(x) = \alpha_2 m(x) + \alpha_4 m^3(x) - \gamma \nabla^2 m(x),$$

so the Euler-Lagrange equation is

$$\gamma \nabla^2 m = \alpha_2 m(x) + \alpha_4 m^3(x)$$

One solution is $m = \text{const.}$, giving the mean field approximation from before. For this reason, mean field theory often means writing down a free energy and working with the Euler-Lagrange equations.

Domain walls

When $T < T_c$, we have two degenerate ground states $m = \pm m_0$. We could have one half of space $x<0$ in the state $m=-m_0$, and the other in $m=m_0$. These are called domains, and the meet at a domain wall.

At the domain wall, let m vary only in one direction. Then

$$\gamma \frac{\partial^2 m}{\partial x} = \alpha_2 m + \alpha_4 m^3.$$

When $m \to \pm m_0$ as $x \to \pm\infty$, this is solved by

$$m = m_0 \tanh\left(\frac{x - X}{W}\right),$$

where X is the position of the domain wall and

$$W = \sqrt{-\frac{2\gamma}{\alpha_2}}$$

its width.